

DreamPrinting: Volumetric Printing Primitives for High-Fidelity 3D Printing

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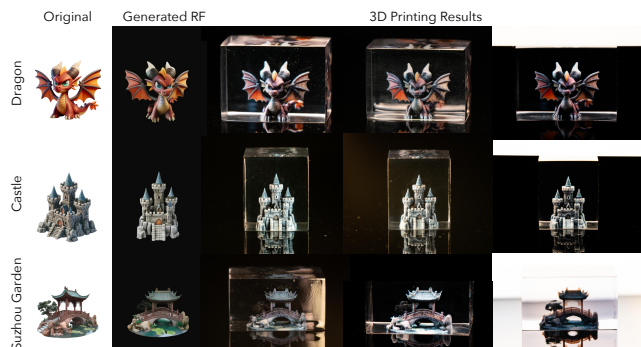


Figure 1: We present *DreamPrinting*, a novel 3D volumetric printing pipeline that enables the reproduction of complex visual effects. Left: A overall demonstration of our printing result. Right: Printings of radiance models generated by TRELLIS[Xiang et al. 2024].

Abstract

We introduce DreamPrinting, a volumetric printing primitives for high-fidelity 3D printing. DreamPrinting tackles this by transforming radiance-based volumetric representations (such as NeRF) into

explicit, material-centric Volumetric Printing Primitives (VPPs) suitable for real-world fabrication. Through cutting-edge 3D printing technology integration, we can accurately restore the radiance field digital asset to centimeter-level accuracy. Through cutting-edge 3D printing technology integration, we can create high-precision centimeter-level objects for restoring radiance field digital assets. (a) Printings of radiance models generated by TRELLIS[Xiang et al. 2024]. From left to right: the prompt image, the generated radiance fields (RF), and the corresponding 3D-printed results. Our approach faithfully captures intricate color details and spatial structures, demonstrating high-fidelity reproduction in physical form.

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SIGGRAPH Emerging Technologies '25, Vancouver, BC, Canada

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ACM ISBN 979-8-4007-1551-8/25/08

<https://doi.org/10.1145/3721257.3734025>

ACM Reference Format:

Yujia Wang, Ruixiang Cao, Teng Xu, Yifei Liu, Dong Zhang, Yiwen Wu, and Jingyi Yu. 2025. DreamPrinting: Volumetric Printing Primitives for High-Fidelity 3D Printing. In *Special Interest Group on Computer Graphics*

and Interactive Techniques Conference Emerging Technologies (SIGGRAPH Emerging Technologies '25), August 10–14, 2025, Vancouver, BC, Canada. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3721257.3734025>

1 Introduction

Three-dimensional (3D) printing has emerged as a transformative technology that enables highly detailed and customized designs in fields such as art, fashion, engineering, and beyond. Modern full-color inkjet 3D printers can print CG models with ultra-fine geometric and textural details, but they often struggle to accurately print volumetric assets that contain internal translucency. The core challenge lies in the absence of suitable primitives that seamlessly integrate volumetric rendering with the physical constraints of 3D printing, including material density, opacity, and light transmission.

Radiance fields can freely combine RGB and density values, whereas inkjet 3D printing is constrained by a limited number of printable pigments, with each voxel assigned a specific pigment. Furthermore, light integration in CG rendering (which aggregates RGB and density along the light path) does not align with real-world pigments, as they exhibit different transmissivities across varying wavelengths.

Inspired by [Abdul-Rahman and Chen 2005; Cao et al. 2025; Sochorová and Jamriška 2021], we propose *DreamPrinting*, a novel volumetric 3D printing pipeline for radiance fields. *DreamPrinting* addresses these challenges by calibrating and modeling the physical properties of pigments and resolving the mismatches between volumetric rendering and volumetric printing. Our approach successfully translates digital volumetric assets, whether hand-crafted, reconstructed, or generated, into physical objects with remarkable detail, as demonstrated in Fig. 1 (left).

2 Method

A *volumetric printing primitive* (VPP) serves as an intermediary that converts a radiance representation, such as a radiance field or radiance volume, into a material-centric 3D printable volume, where each voxel is assigned a specific pigment composition. By sampling the radiance representation at the grid points corresponding to each VPP, we extract the RGB color values and density.

To accurately convert volumetric rendering primitives (VRPs) with RGB color and density into volumetric printing primitives (VPPs), we designed a theoretical framework. First, we assume that each voxel can contain multiple pigments mixed at arbitrary concentrations, which transforms the problem from a discrete optimization challenge into a continuous one. Next, we build upon the Kubelka-Munk (K-M) model to establish a mapping from pigment concentrations to RGB color and density values, which accurately simulates the optical properties of pigment mixtures within a small local region.

To achieve the inverse mapping from RGB color and density to K-M model parameters, we solve the inverse K-M problem. However, due to the transcendental nature of these equations, obtaining an analytical solution is infeasible. To address this, we extend the method proposed in [Sochorová and Jamriška 2021] for rapidly mapping opaque pigment concentrations to RGB values. Through pigment calibration, we establish a printable color space, enabling

efficient lookup of pigment concentration combinations required to achieve a target color.

Once the optimal pigment concentrations are determined, we apply a 3D stochastic halftoning technique inspired by 2D halftoning. This method performs random sampling based on the computed pigment concentrations, ultimately yielding the final pigment assignments at each voxel. For further implementation details, please refer to our supplementary technical paper.

3 Results

We conducted experiments using a Stratasys J850 Prime [Stratasys 2025], with six primary pigments (Cyan, Magenta, Yellow, Black, White, and Clear). Figure 1(left) highlights several 3D-printed examples. Compared to traditional surface-based color 3D prints, *DreamPrinting* excels in capturing subtle interior details such as translucent fur and clouds.

In Figure 1 (right), we showcase radiance fields generated by TRELIS [Xiang et al. 2024]—a text/image-to-3D system—alongside their physical counterparts. Complex structures are rendered with faithful internal coloration and optical depth cues. The prints range from 2-3 cm in their longest dimension and retain fine features.

4 Conclusion

DreamPrinting demonstrates a robust pipeline that unifies volumetric rendering, pigment calibration, and high-resolution inkjet printing. This advance expands the scope of 3D printing beyond surface-color reproduction, enabling the faithful fabrication of volumetric assets from both real-world captures and generative models. By resolving mismatches in optical integration and limited pigment availability, *DreamPrinting* lays the foundation for a new era of tangible, high-fidelity volumetric prints—empowering artists, researchers, and engineers to transform intricate digital radiance fields into stunning physical artifacts. Future work could explore additional pigments, faster optimization strategies, and adaptive halftoning for even larger or more complex volumetric scenes.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China under W2431046, and by the MoE Key Lab of Intelligent Perception and Human-Machine Collaboration (ShanghaiTech University), the Shanghai Frontiers Science Center of Human-centered Artificial Intelligence, and the HPC Platform of ShanghaiTech University.

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